Practice Assignment: Simple Input and Output

Overview

In this assignment, you will create a small C++ program that stores and displays numeric student and course information. You will organize the variables using two namespaces: `student` and `course`.

Both namespaces will contain a variable named `id`. This will demonstrate how namespaces allow the same variable name to be used for different purposes without creating a naming conflict.

> **Important:** Complete this assignment using only the C++ topics introduced so far. Do not use strings, loops, conditional statements, or additional functions.

Learning Objectives

By completing this assignment, you will practice how to:

- Create, compile, and run a C++ program in GitHub Codespaces.
- Navigate folders and inspect your location using Unix terminal commands.
- Declare integer variables.
- Create and use two custom namespaces.
- Access namespace members with the scope resolution operator (`::`).
- Read numeric input with `cin`.
- Display formatted output with `cout`.

GitHub Codespaces Workflow

Complete the following steps in the GitHub Codespaces terminal. The commands shown are part of the required workflow.

1. Check your current location

```
pwd
```

`pwd` means **print working directory**. It displays the path of your current directory.

2. Create an assignment directory

```
mkdir simple_input_output
```

`mkdir` means **make directory**. It creates a new folder named `simple_input_output`.

3. Enter the new directory

```
cd simple_input_output
```

`cd` means **change directory**. It moves you into the specified directory.

4. Confirm your new location

```
pwd
```

The displayed path should now end with `simple_input_output`.

5. View the directory contents

```
ls
```

`ls` means **list**. It displays the contents of the current directory. The directory may be empty at this point.

6. Create and open the source file

```
code registration.cpp
```

The `code` command creates the file if it does not exist and opens it in the editor. Write your program (see the task below) in `registration.cpp`, and remember to save it.

7. Check that the file exists

```
ls
```

You should see `registration.cpp` in the list.

8. Clear the terminal

```
clear
```

`clear` clears the terminal window so that the next commands and their output are easier to read.

9. Compile the program

```
g++ registration.cpp
```

`g++` compiles the C++ source file.

If the compiler reports errors, read the messages, correct the source code, save the file, and compile again.

10. Run the program

```
./out
```

The `./` tells the terminal to run the executable file located in the current directory.

11. Test the program

Enter values when prompted. Confirm that the summary displays the same four values you entered. Run the program more than once with different numbers.

Assignment

Write a C++ program that collects numeric information about one student and one course.

Create these two namespaces **above** `main()`:

`student` namespace

The `student` namespace must contain:

- `int id` — the student's identification number

`course` namespace

The `course` namespace must contain:

- `int id` — the course identification number

Required Program Behavior

Your program must:

1. Ask the user to enter a student ID.
2. Store the value in `student::id`.
3. Ask the user to enter a course ID.
4. Store the value in `course::id`.
5. Display a clearly labeled registration summary containing all four values.

Use the scope resolution operator (`::`) whenever you access a variable inside either namespace.

Sample Interaction

Your wording and spacing may differ slightly, but the program should behave like this:

```
Enter student ID: 12345
Enter course ID: 3530

--- Registration Information ---
Student ID: 12345
Course ID: 3530
```